**Lectures 19 – 21**

Math 4741: Mathematical Analysis

Chapter 8: Hidden Markov Models

References

Texts: Markov Processes for Stochastic Modeling – Oliver Ibe (Chapter 14)

# Hidden Markov Models (HMMs)

HMM Basics:

An HMM is usually defined as a 5-tuple $(S, \Omega, P, \Phi, \pi)$

$S =$

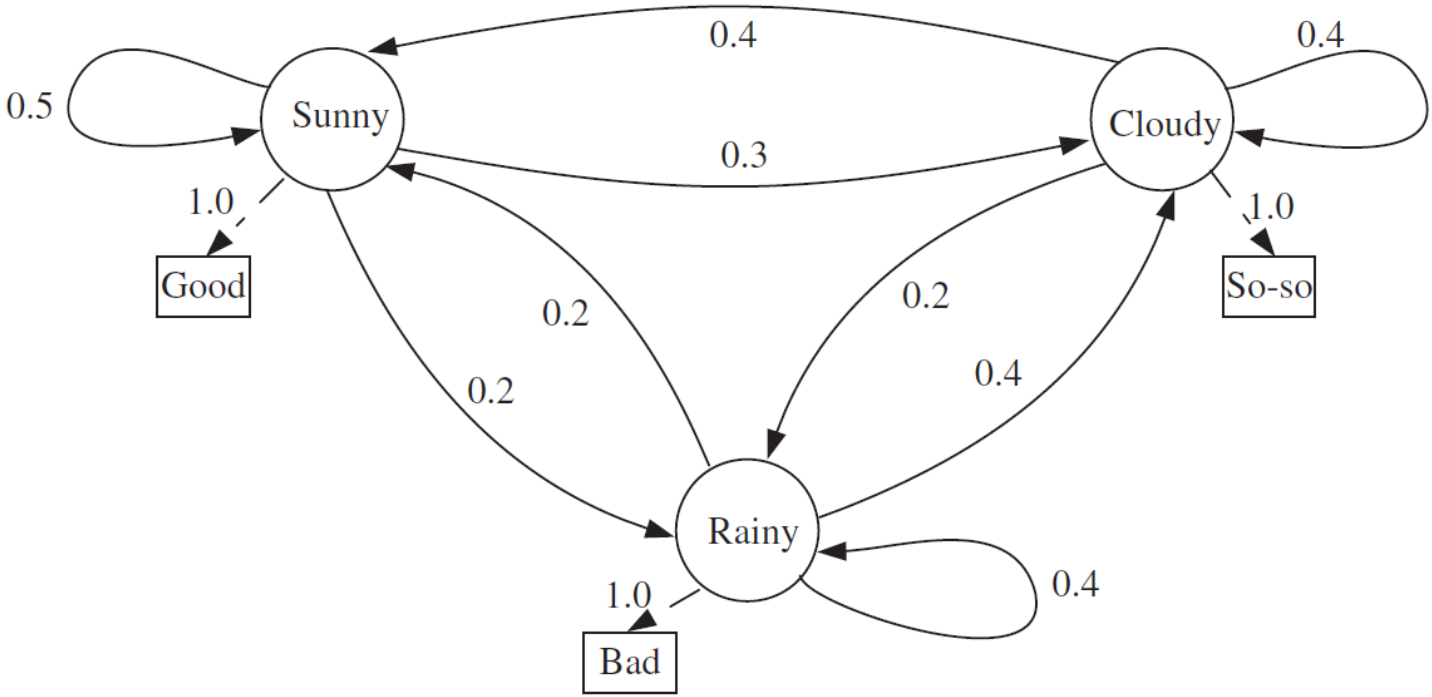
$\Omega =$

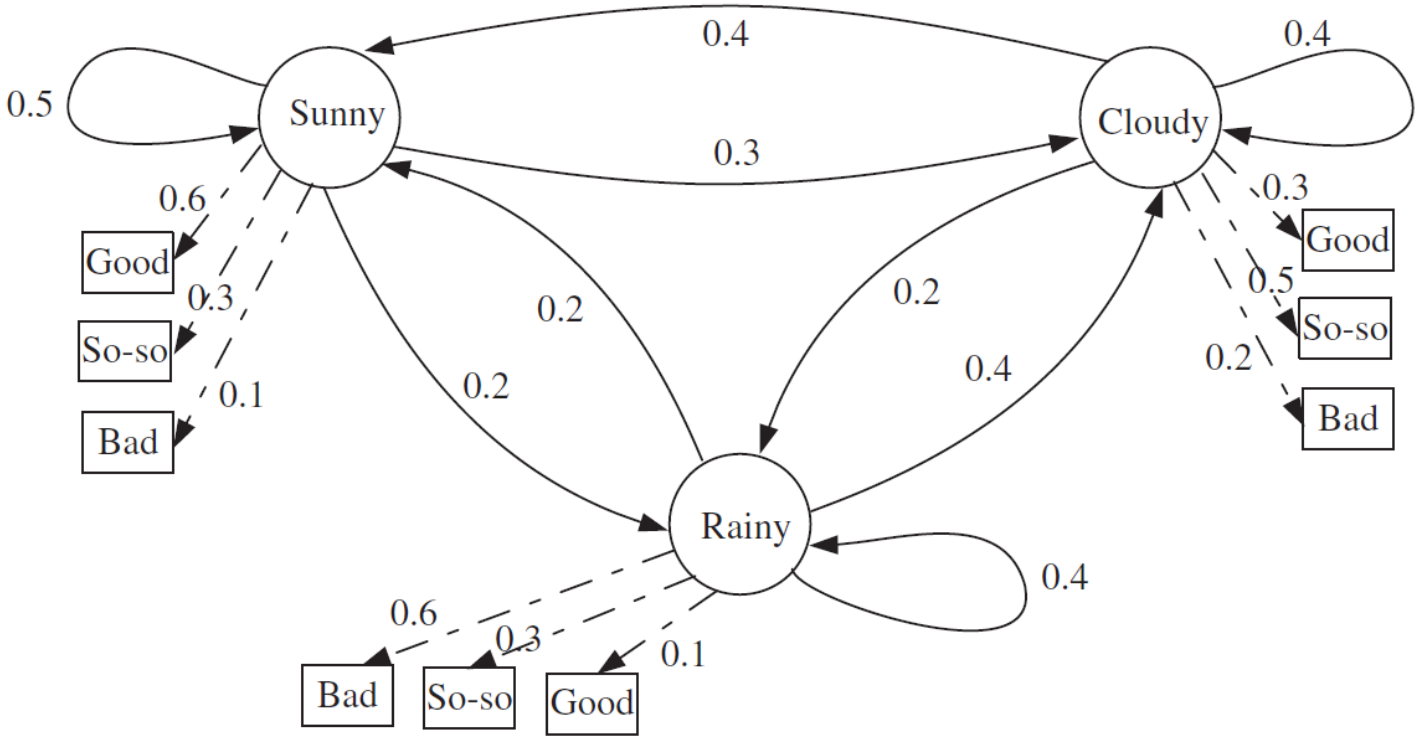
$P =$

$\Phi =$

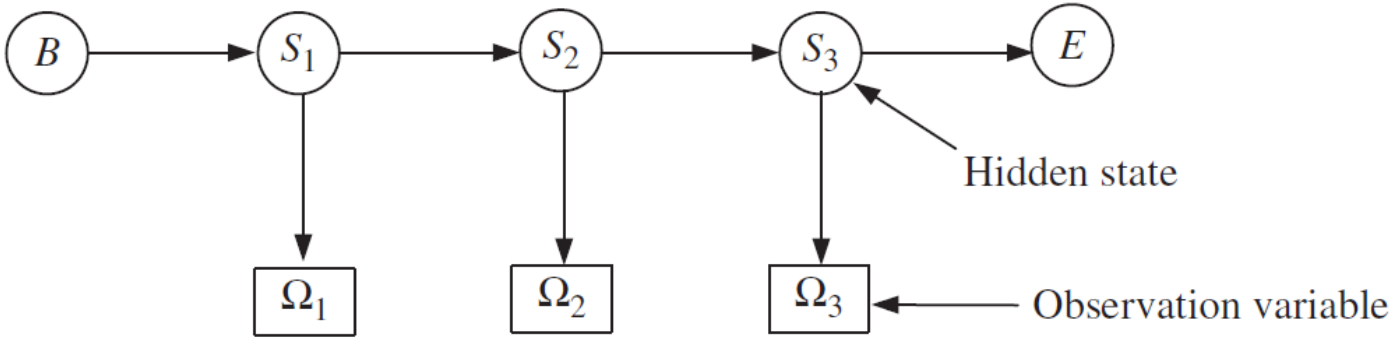
$\pi =$

Parameters of an HMM


0.4
0.4
0.5
Sunny
Cloudy
0.3
1.0
1.0
Good
So-so
0.2
0.2
0.2
0.4
Rainy
0.4
1.0
Bad


Sunny
Cloudy
Rainy
0.5
0.4
0.3
0.4
0.6
0.3
0.1
0.2
0.2
0.2
0.4
0.3
0.5
0.2
0.4
0.6
0.3
0.1
Good
So-so
Bad
Good
So-so
Bad
Bad
So-so
Good

General Structure of an HMM



HMM Assumptions

1. Markov assumptions

2. Stationary assumptions

3. Observation independence assumptions

Three Fundamental Problem

1. *The evaluation problem*: Given a model $\lambda = (P, \Phi, \pi)$ and an observation sequence $O = v_1, v_2, \ldots, v_T$ of length $T$, where $v_i \in \Omega$, how do we efficiently compute the probability that the model generated the observation sequence; that is, what is $P[O|\lambda]$?

2. *The decoding problem*: Given a model $\lambda = (P, \Phi, \pi)$, what is the most likely sequence of hidden states that could have generated a given observation sequence? Thus, we would like to find $Q^* = \arg\max_Q P[Q, O|\lambda]$, where $Q$ is the hidden state sequence, as defined earlier.

3. *The learning problem*: Given a set of observation sequences, find the HMM that best explains the observation sequences; that is, find the values for $\lambda$ that maximize $P[O|\lambda]$ or $\lambda^* = \arg\max_\lambda P[O|\lambda]$. Stated differently, the problem is to estimate the most likely HMM parameters for a given observation sequence.

Solution Methods

1. Evaluation Problem: Forward algorithm and backward algorithm

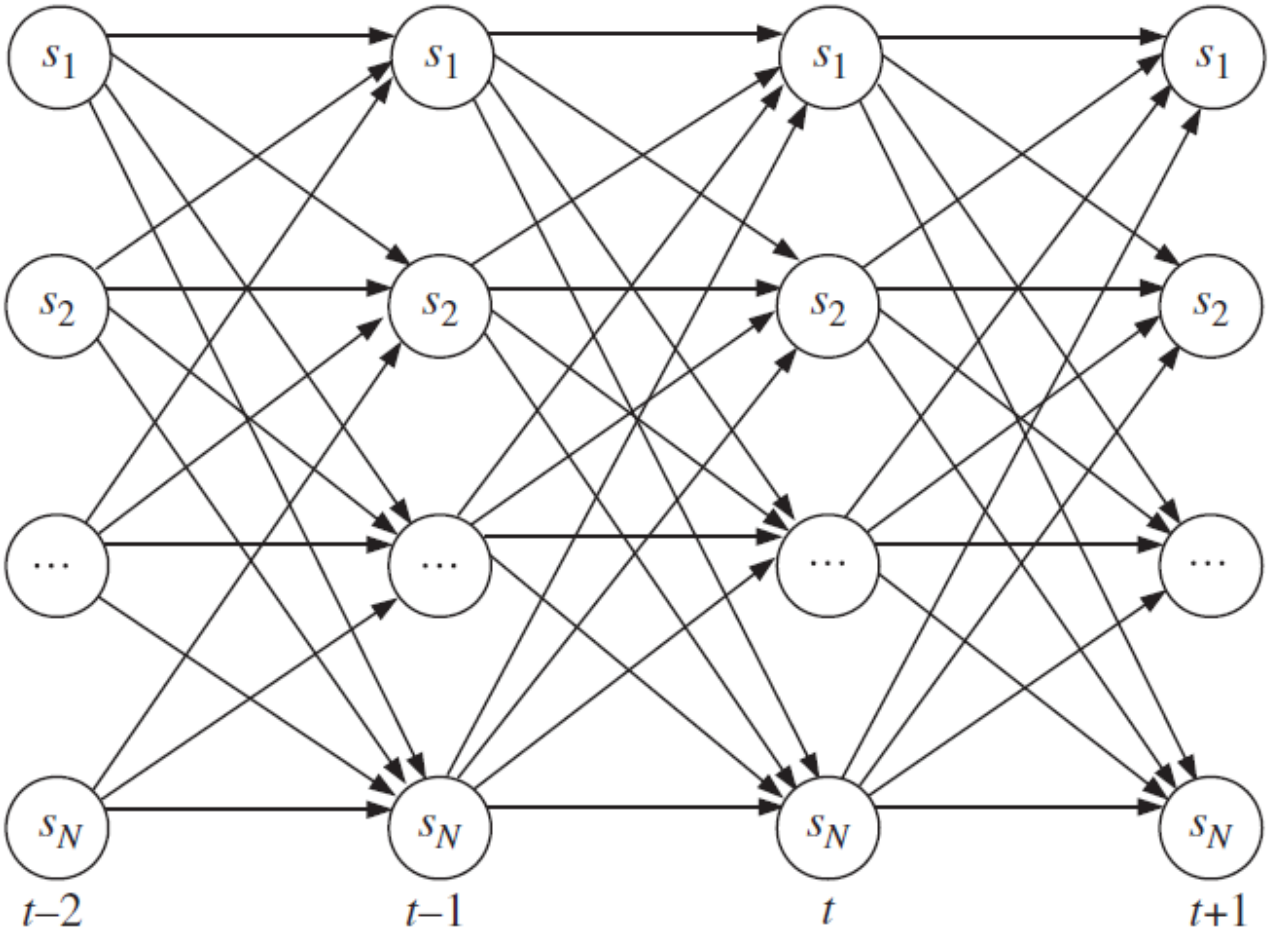
2. Decoding problem: Viterbi algorithm

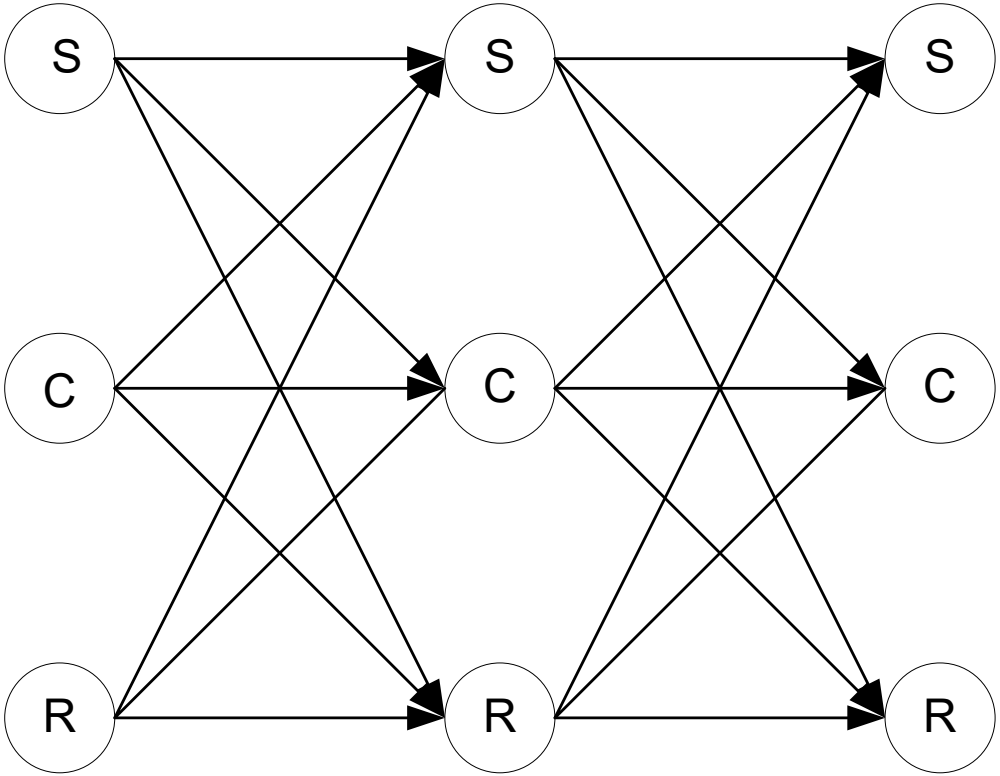
3. Learning Problem: Baum-Welch algorithm

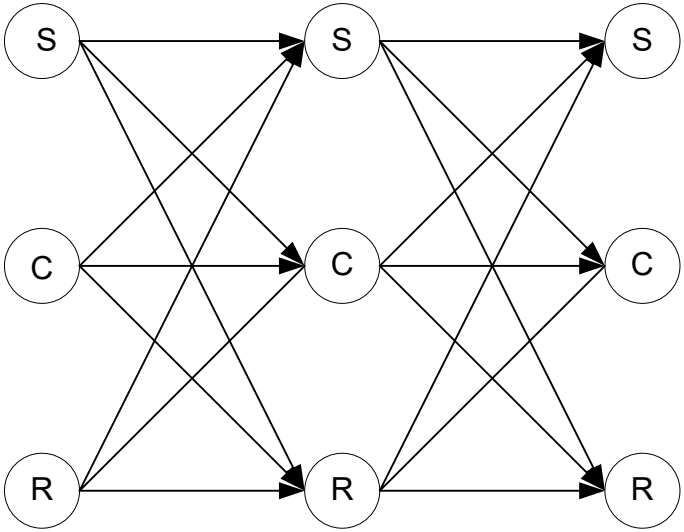
The Evaluation method

The forward Algorithm

A forward probability variable $\alpha_t(i)$ is used:


$s_1$
$s_2$
…
$s_N$
$t-2$
$t-1$
$t$
$t+1$


S
C
R
S
C
R
S
C
R


S
C
R
S
C
R
S
C
R

## The forward algorithm

**1.** Initialization:

$$\alpha_1(i) = \pi_i \phi_i(o_1) \quad 1 \le i \le N$$

**2.** Induction:

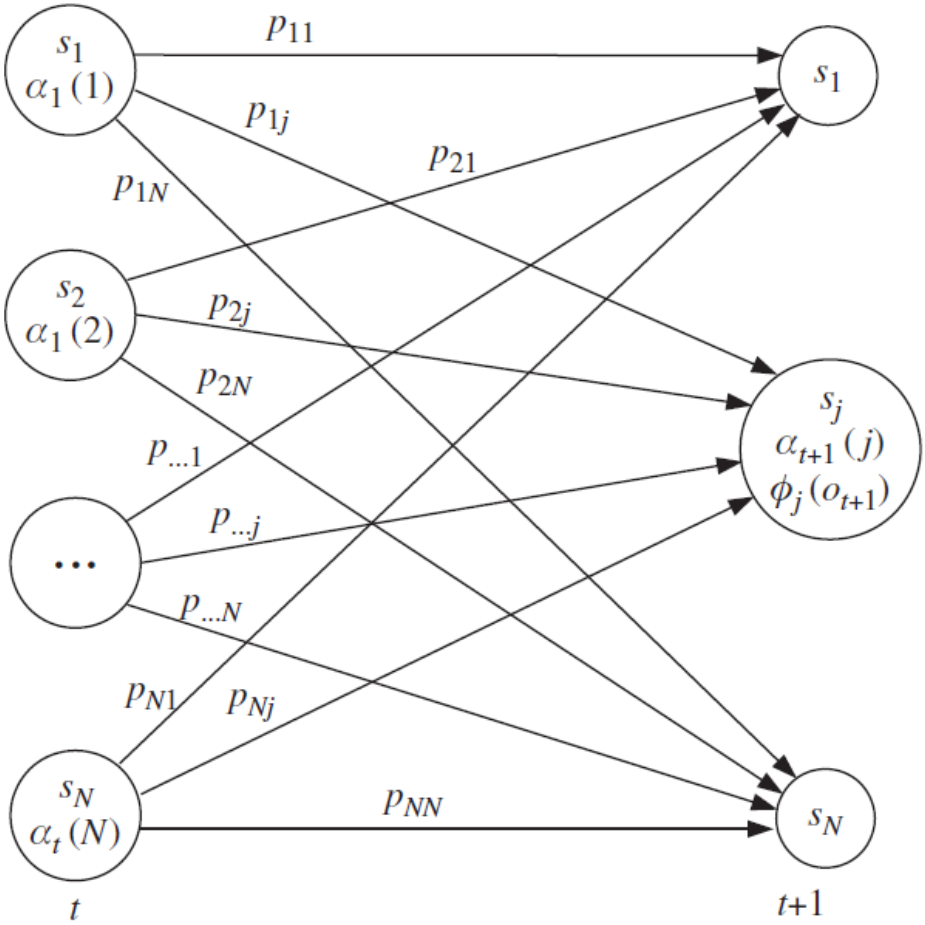
$$\alpha_{t+1}(j) = \left\{ \sum_{i=1}^{N} p_{ij} \alpha_t(i) \right\} \phi_j(o_{t+1}) \quad 1 \le t \le T-1, \quad 1 \le j \le N$$

**3.** Update time: Set $t = t + 1$. If $t < T$, go to step 2; otherwise, go to step 4.

**4.** Termination:

$$P[O|\lambda] = \sum_{i=1}^{N} \alpha_T(i) = \sum_{i=1}^{N} P[O, q_T = s_i|\lambda]$$

The forward algorithm requires $N(N+1)(T-1) + N$ multiplications and $N(N-1)(T-1)$ additions, giving a complexity on the order of $N^2T$ rather than $2TN^T$.


$s_1$
$\alpha_1(1)$
$s_2$
$\alpha_1(2)$
...
$s_N$
$\alpha_t(N)$
$t$
$p_{11}$
$p_{1j}$
$p_{1N}$
$p_{21}$
$p_{2j}$
$p_{2N}$
$p_{...1}$
$p_{...j}$
$p_{...N}$
$p_{N1}$
$p_{Nj}$
$p_{NN}$
$s_1$
$s_j$
$\alpha_{t+1}(j)$
$\phi_j(o_{t+1})$
$s_N$
$t+1$

## Example 14.1

Consider Bill's mood change problem illustrated in Figure 13.2. Assume that we observed Bill in the following sequence of moods: *Good*, *Good*, *So-so*, *Bad*, *Bad*. We are required to find the probability that the model generated such a sequence of moods using the forward algorithm.

### Solution

Let $S$ denotes Sunny state, $C$ denotes Cloudy, and $R$ denotes Rainy. Similarly, let $G$ denotes Good mood, $SS$ denotes So-so mood, and $B$ denotes Bad mood. We assume that the process is equally likely to start in any state. That is, we assume that $\pi_S = \pi_C = \pi_R = 1/3$. Also, we have that $T = 5$. Then the initialization step becomes

# The Initialization step

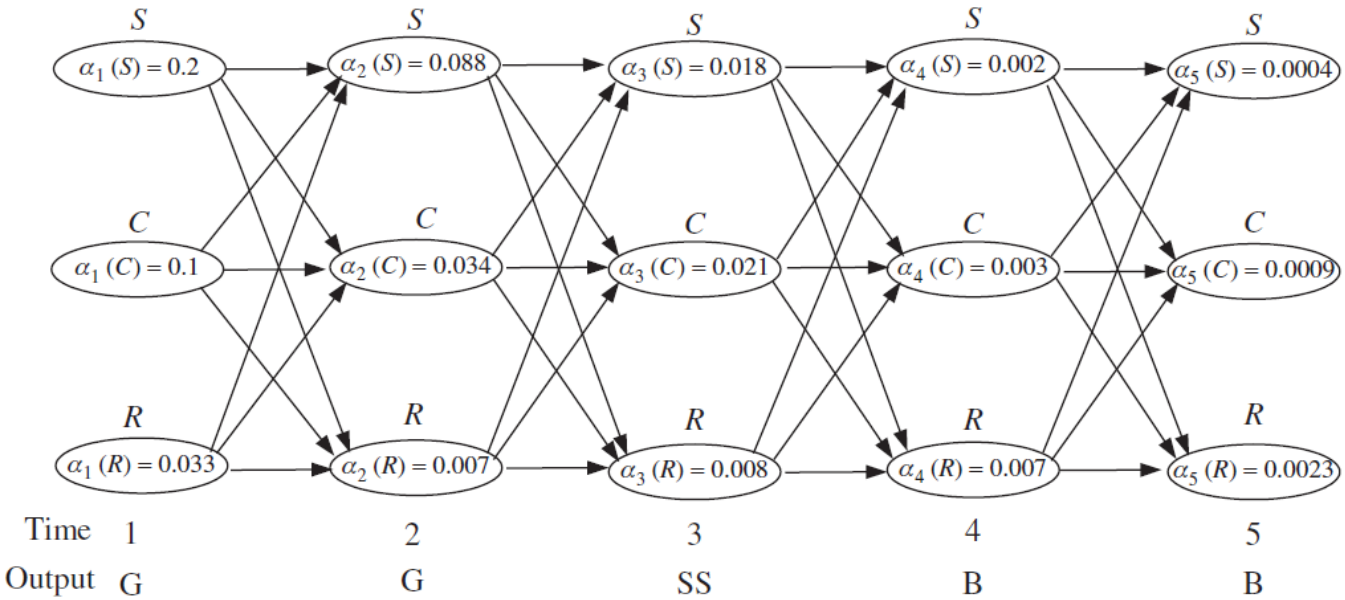
The induction step for $t = 2$ is given by

$$\alpha_2(j) = \left\{ \sum_{i=1}^{N} p_{ij}\alpha_1(i) \right\} \phi_j(o_2) = \left\{ \sum_{i=1}^{N} p_{ij}\alpha_1(i) \right\} \phi_j(G)$$

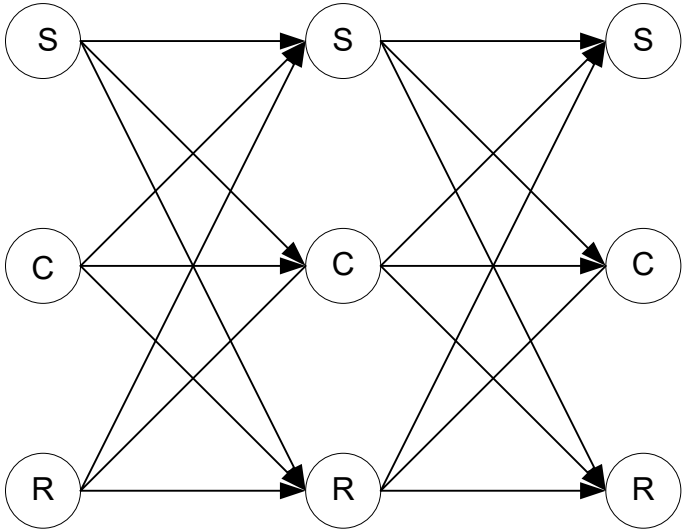
$$\begin{aligned}
\alpha_2(S) &= \{p_{SS}\alpha_1(S) + p_{CS}\alpha_1(C) + p_{RS}\alpha_1(R)\}\phi_S(G) \\
&= \{(0.5)(0.2) + (0.4)(0.1) + (0.2)(0.033)\}(0.6) \\
&= 0.088 \\
\alpha_2(C) &= \{p_{SC}\alpha_1(S) + p_{CC}\alpha_1(C) + p_{RC}\alpha_1(R)\}\phi_C(G) \\
&= \{(0.3)(0.2) + (0.4)(0.1) + (0.4)(0.033)\}(0.3) \\
&= 0.034 \\
\alpha_2(R) &= \{p_{SR}\alpha_1(S) + p_{CR}\alpha_1(C) + p_{RR}\alpha_1(R)\}\phi_R(G) \\
&= \{(0.2)(0.2) + (0.2)(0.1) + (0.4)(0.033)\}(0.1) \\
&= 0.007
\end{aligned}$$

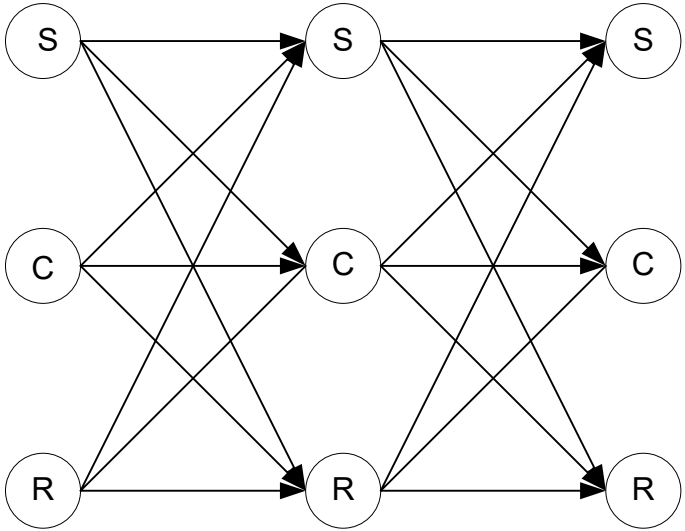
Thus, at the termination of the algorithm, we obtain the solution as

$$P[O = G, G, SS, B, B|\lambda] = \sum_{i=1}^{N} \alpha_T(i) = \alpha_T(S) + \alpha_T(C) + \alpha_T(R) = 0.0036$$


S
C
R
$\alpha_1(S) = 0.2$
$\alpha_1(C) = 0.1$
$\alpha_1(R) = 0.033$
$\alpha_2(S) = 0.088$
$\alpha_2(C) = 0.034$
$\alpha_2(R) = 0.007$
$\alpha_3(S) = 0.018$
$\alpha_3(C) = 0.021$
$\alpha_3(R) = 0.008$
$\alpha_4(S) = 0.002$
$\alpha_4(C) = 0.003$
$\alpha_4(R) = 0.007$
$\alpha_5(S) = 0.0004$
$\alpha_5(C) = 0.0009$
$\alpha_5(R) = 0.0023$
Time 1 2 3 4 5
Output G G SS B B

# The Backward Algorithm


S
C
R
S
C
R
S
C
R


S
C
R
S
C
R
S
C
R

Initialization

Induction

Update

Termination